

3.1.1. Numpy Array Operations

05:48

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `rows` and `cols`, representing the number of rows and the number of columns.
- The next `rows` lines contain `cols` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Code:

```
import numpy as np

rows,cols = map(int,input().split())

elements = []

for _ in range(rows):

    elements.extend(map(int,input().split()))

array=np.array(elements).reshape(rows,cols)

print(array)
```

```
print(array.ndim)
```

```
print(array.shape)
```

```
print(array.size)
```

Explorer

Average time
0.035 s
35.08 ms

Maximum time
0.089 s
89.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 10 out of 10 hidden test case(s) passed

✓ Test case 1 89 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[·1·2·3·4]]	[[·1·2·3·4]]
·[·5·6·7·8]	·[·5·6·7·8]
·[·9·10·11·12]	·[·9·10·11·12]
2	2
(3, 4)	(3, 4)
12	12

✓ Test case 2 20 ms Debug

Expected output	Actual output
0 0	0 0
[]	[]
2	2
(0, -0)	(0, -0)
0	0